

Now we lay down Prop. VI's apparatus: QT perpendicular to SP and LR parallel to SP , meeting the tangent at R and Z and the circle at L . The similar triangles ZQR , ZTP , ZPA give RP^2 (that is $QR \cdot L$) to QT^2 as AV^2 to PV^2 , so multiplying by SP^2/QR and letting Q coincide with P collapses the construction into the clean chord-cube. Substituting into Prop. VI yields $F \propto \frac{1}{SP^2 \cdot PV^3}$; and in the special case where S sits at O , the chord PV becomes the diameter and the law reduces to the pure inverse-square $1/SP^2$. *Q.E.D.*